0.1 Riemann-Stieltjes Integral

Definition 0.1.1. A finite subset $P \subset [a,b]$ containing a, b is called a *Partition* of $[a.b]$.
Conventionally, denote $P = \{x_0, x_1, \dots, x_n\}$ where $a = x_0 < x_1 < \cdots < x_n = b$.
The set of Partitions of $[a,b]$ is denoted as $\mathcal{P}[a,b]$.

Definition 0.1.2. Let $f : [a,b] \to \mathbb{R}$ be a bounded map, and $\alpha : [a,b] \to \mathbb{R}$ be a monotone-increasing.
Let a partition $P \in \mathcal{P}[a,b]$ be given, and denote $P = \{x_0, x_1, \dots, x_n\}$. For each $1 \leq i \leq n$, define

$$\Delta\alpha_i \overset{\text{def}}{=} \alpha(x_i) - \alpha(x_{i-1}) \underset{\alpha \text{ mono.}}{\geq} 0$$

and

$$M_i \overset{\text{def}}{=} \sup\{f(t) \mid x_{i-1} \leq t \leq x_i\} \text{ and } m_i \overset{\text{def}}{=} \inf\{f(t) \mid x_{i-1} \leq t \leq x_i\}$$

And, define *Upper Sum* and *Lower Sum* as:

$$U(f,P,\alpha) \overset{\text{def}}{=} \sum_{i=1}^{n} M_i \cdot [\alpha(x_i) - \alpha(x_{i-1})] \text{ and } L(f,P,\alpha) \overset{\text{def}}{=} \sum_{i=1}^{n} m_i \cdot [\alpha(x_i) - \alpha(x_{i-1})]$$

Now, define *Upper integral* and *Lower integral* as:

$$\overline{\int_a^b} f d\alpha \overset{\text{def}}{=} \inf_{P \in \mathcal{P}[a,b]} U(f,P,\alpha) \text{ and } \underline{\int_a^b} f d\alpha \overset{\text{def}}{=} \sup_{P \in \mathcal{P}[a,b]} L(f,P,\alpha)$$

We said that f is *Riemann-Stieltjes Integrable* respect to α if:

$$\overline{\int_a^b} f d\alpha = \underline{\int_a^b} f d\alpha$$

The set of Riemann-Stieltjes Integrable functions respect to α as $\mathcal{R}(\alpha)$.

Lemma 0.1.3. Let $f : [a,b] \to \mathbb{R}$ be a bounded map, and $\alpha : [a,b] \to \mathbb{R}$ be a monotone-increasing.
For any partitions $P, Q \subset [a,b]$, and a refinement $P^* \supset P$,

$$\underline{\int_a^b} f d\alpha \leq L(f,P,\alpha) \leq L(f,P^*,\alpha) \leq U(f,P^*,\alpha) \leq U(f,P,\alpha) \leq \overline{\int_a^b} f d\alpha$$

and

$$L(f,P,\alpha) \leq U(f,Q,\alpha)$$

0.1.1 Properties of Riemann-Stieltjes Integral

Lemma 0.1.4. Let $f : [a, b] \to \mathbb{R}$ be a bounded map, and $\alpha : [a, b] \to \mathbb{R}$ be a monotone-increasing.
If $f, g \in \mathcal{R}(\alpha)$ and $c \in \mathbb{R}$, then $f + g,\ cf \in \mathcal{R}(\alpha)$ with

$$\int_a^b (f+g) d\alpha = \int_a^b f d\alpha + \int_a^b g d\alpha$$
$$\int_a^b (cf) d\alpha = c \int_a^b f d\alpha$$

Lemma 0.1.5. Let $f : [a, b] \to \mathbb{R}$ be a bounded map, and $\alpha : [a, b] \to \mathbb{R}$ be a monotone-increasing.
If f is Riemann-Stieltjes Integrable respect to α on $[a, b]$ and $a < c < b$,
then f is Riemann-Stieltjes Integrable respect to $[a, c]$ and $[c, b]$ with

$$\int_a^b f d\alpha = \int_a^c f d\alpha + \int_c^b f d\alpha$$

Lemma 0.1.6. Let $f : [a, b] \to \mathbb{R}$ be a bounded map, and $\alpha, \beta : [a, b] \to \mathbb{R}$ be monotone-increasing.
If $f \in \mathcal{R}(\alpha)$, $f \in \mathcal{R}(\beta)$ and $c > 0$, then $f \in \mathcal{R}(\alpha + \beta)$ and $f \in \mathcal{R}(c\alpha)$ with

$$\int_a^b f d(\alpha + \beta) = \int_a^b f d\alpha + \int_a^b f d\beta$$
$$\int_a^b f d(c\alpha) = c \int_a^b f d\alpha$$

Theorem 0.1.7. Let $f : [a, b] \to \mathbb{R}$ be a bounded map, and $\alpha : [a, b] \to \mathbb{R}$ be monotone-increasing. TFAE:

1. f is Riemann-Stieltjes Integrable respect to α on $[a, b]$.
2. For any $\varepsilon > 0$, there exists a Partition $P \in \mathcal{P}[a, b]$ such that $U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon$.
3. For some $A \in \mathbb{R}$, for any $\varepsilon > 0$, there exists $P_0 \in \mathcal{P}[a, b]$ such that

$$\forall P \in \mathcal{P}[a, b],\ \ P \supseteq P_0 \implies |S(f, P, \alpha) - A| < \varepsilon$$

where Riemann-Stieltjes Sum $S(f, P, \alpha) \overset{\text{def}}{=} \sum_{i=1}^{n} f(t_i)[\alpha(x_i) - \alpha(x_{i-1})]$, for some $t_i \in [x_{i-1}, x_i]$.

0.1.2 Riemann-Stieltjes Integrable

> **Theorem 0.1.8.** If $f:[a,b]\to\mathbb{R}$ is continuous, then $f\in\mathcal{R}(\alpha)$ for any monotone increasing $\alpha:[a,b]\to\mathbb{R}$.

> **Theorem 0.1.9.** If $f:[a,b]\to\mathbb{R}$ is monotone and $\alpha:[a,b]\to\mathbb{R}$ is monotone increasing continuous, then $f\in\mathcal{R}(\alpha)$.

0.1.3 Riemann-Integral

Definition 0.1.10. Let $\mathcal{P}[a,b]$ be a set of partitions. Define a *Norm* as:

$$\|\cdot\| : \mathcal{P}[a,b] \to \mathbb{R} : P \mapsto \max\{|x_i - x_{i-1}| \mid i = 1, 2, \ldots, n\} \text{ where } P = \{x_0, x_1, \ldots, x_n\}$$

Theorem 0.1.11. Let $f : [a,b] \to \mathbb{R}$ and $A \in \mathbb{R}$, TFAE:

1. f is Riemann-Integral with $\displaystyle\int_a^b f = A$.
2. For any $\varepsilon > 0$, there exists $\delta > 0$ such that $\forall P \in \mathcal{P}[a,b],\ \|P\| < \delta \implies |R(f,P) - A| < \varepsilon$.
3. For any $\varepsilon > 0$, there exists $\delta > 0$ such that $\forall P \in \mathcal{P}[a,b],\ P \supseteq P_0 \implies |R(f,P) - A| < \varepsilon$.

The statement 2) can be written as: $\displaystyle\lim_{\|P\| \to 0} R(f,P) = A$.

0.1.4 Bounded Variation Function

Definition 0.1.12. Let $f : [a, b] \to \mathbb{R}$ be a bounded function and $P = \{x_0, x_1, \dots, x_n\} \in \mathcal{P}[a, b]$.
Define a *Variation* of f with respects to P as:

$$V_a^b(f, P) \stackrel{\text{def}}{=} \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})|$$

If the set $\{V_a^b(f, P) \mid P \in \mathcal{P}[a, b]\}$ is bounded above, then f is called *Bounded Variation Function*.
In this case, define a *Total Variation* of f as:

$$V_a^b(f) \stackrel{\text{def}}{=} \sup\{V_a^b(f, P) \mid P \in \mathcal{P}[a, b]\}$$